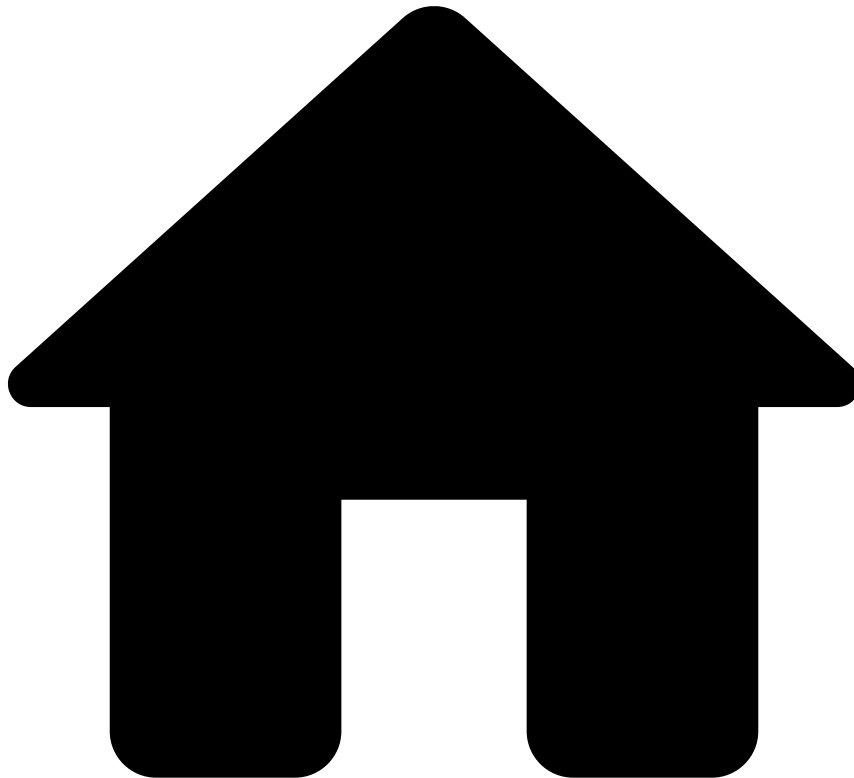


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Transfers - Fraud Rules

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Description

Fraud Rules look for common fraud patterns in transfers. If triggered, some rules included in this rules project mark the event as `high` risk, which is then marked as `decline` by the *Decision Rules* (i.e. if the `enforced_risk_level` does not override the risk outcome). Other rules mark the event as `medium` risk, and then the *Decision Rules* can trigger an alert. In the case of generating an alert, the event can be further investigated by fraud analysts in Case Manager.

Rules Project Metadata

- *Target Variable:* `risk_level`

- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-FR1

Comment

Number of transactions per sender in 90 minutes exceeds threshold

Actions

- [Transfers-FR1]

Variables

```
min_count_trx_sender_90min_th =
transfers__rules_thresholds["min_count_trx_sender_90min_th"].threshold;
```

Condition

```
event.count_sender_90min >= min_count_trx_sender_90min_th
```

Explanation

[Transfers-FR1] - High Risk - Count of transactions by this Sender ({event.count_sender_90min}) in the last 90min is above {localvar.min_count_trx_sender_90min_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR2

Comment

Number of transactions per sender in 24 hours exceeds threshold

Actions

- [Transfers-FR2]

Variables

```
min_count_trx_sender_24h_th =
transfers__rules_thresholds['min_count_trx_sender_24h_th'].threshold;
```

Condition

```
event.count_sender_24h >= min_count_trx_sender_24h_th
```

Explanation

[Transfers-FR2] - High Risk - Count of transactions by this Sender ({event.count_sender_24h}) in the last 24h is above {localvar.min_count_trx_sender_24h_th}.

Mark As

The outcome of this rule is: **high**

Transfers-FR3

Comment

Daily limit on the number of different country codes per sender

Actions

- [Transfers-FR3]

Variables

```
min_distinct_country_code_per_sender_24h_th
=transfers__rules_thresholds["min_distinct_country_code_per_sender_24h_th"].threshold;
```

Condition

```
event.distinct_country_per_sender_24 >= min_distinct_country_code_per_sender_24h_th
```

Explanation

[Transfers-FR3] - High Risk - Count of distinct country codes ({event.distinct_country_per_sender_24}) per Sender in 24h is above {localvar.min_distinct_country_code_per_sender_24h_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR5

Comment

Number of daily transactions for a dormant account exceeds threshold and transaction amount exceeds percentage threshold of account balance.

Actions

- [Transfers-FR5]

Variables

```
min_count_trx_sender_dormant_th =
transfers__rules_thresholds["min_count_trx_sender_dormant_th"].threshold;
min_amount_account_balance_percentage_th =
transfers__rules_thresholds["min_amount_account_balance_percentage_th"].threshold;

account_age_months = ((event.created_at ?? 0) - (event.sender_account_open_date ?? 0)) /
2678400000.0; // 1000 * 60 * 60 * 24 * 31

amount_account_balance_percentage = Maths.safeDivision((event.sender_transaction_amount ?? 0),
(event.sender_account_balance ?? 0));
```

Condition

```
event.count_activity_3m_delay_24h == 0
and
event.count_sender_24h >= min_count_trx_sender_dormant_th
and
account_age_months >= 6
and
amount_account_balance_percentage >= min_amount_account_balance_percentage_th
```

Explanation

[Transfers-FR5] - High Risk - This sender has no associated transactions in the last 3 months, but in the last 24 hours it suddenly has {event.count_sender_24h} associated transactions.

Mark As

The outcome of this rule is: **high**

Transfers-FR6

Comment

Daily limit on international transactions over 3K per sender

Actions

- [Transfers-FR6]

Variables

```
min_count_trx_sender_cross_border_amount_3k_24h_th =  
transfers__rules_thresholds["min_count_trx_sender_cross_border_amount_3k_24h_th"].threshold;
```

Condition

```
event.count_sender_cross_border_amsr >= min_count_trx_sender_cross_border_amount_3k_24h_th
```

Explanation

[Transfers-FR6] - High Risk - Count of international transactions by this Sender ({event.count_sender_cross_border_amsr}) in the last 24h in is above {localvar.min_count_trx_sender_cross_border_amount_3k_24h_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR7

Comment

Monthly limit on international transactions over 3K per sender

Actions

- [Transfers-FR7]

Variables

```
min_count_trx_sender_cross_border_amount_3k_1m_th =  
transfers__rules_thresholds["min_count_trx_sender_cross_border_amount_3k_1m_th"].threshold;
```

Condition

```
event.count_sender_cross_border_amou >= min_count_trx_sender_cross_border_amount_3k_1m_th
```

Explanation

[Transfers-FR7] - High Risk - Count of international transactions by this Sender ({event.count_sender_cross_border_amou}) in the last month in is above {localvar.min_count_trx_sender_cross_border_amount_3k_1m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR11

Comment

Daily limit on different receivers per sender

Actions

- [Transfers-FR11]

Variables

```
min_distinct_receiver_per_sender_24h_th =transfers__rules_thresholds  
["min_distinct_receiver_per_sender_24h_th"].threshold;
```

Condition

```
event.distinct_receiver_per_sender_2 >=min_distinct_receiver_per_sender_24h_th
```

Explanation

[Transfers-FR11] - High Risk - Count of distinct receivers ({event.distinct_receiver_per_sender_2}) per Sender in 24 hours is above {localvar.min_distinct_receiver_per_sender_24h_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR13

Comment

Daily limit on different receivers with the same name per sender

Actions

- [Transfers-FR13]

Variables

```
min_distinct_receiver_same_name_per_sender_24h_th  
=transfers__rules_thresholds["min_distinct_receiver_same_name_per_sender_24h_th"].threshold;
```

Condition

```
event.distinct_receiver_same_name_pe >= min_distinct_receiver_same_name_per_sender_24h_th
```

Explanation

[Transfers-FR13] - High Risk - The number of distinct Receivers with the same name that received transfers from this Sender ({event.distinct_receiver_same_name_pe}) in the last 24h is above {localvar.min_distinct_receiver_same_name_per_sender_24h_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR15

Comment

Limit on transfers outside sender's usual hours and is to a new receiver (6 months)

Actions

- [Transfers-FR15]

Variables

```
max_logins_time_of_day_th =
transfers__rules_thresholds['max_account_activity_time_of_day_3m_th'].threshold;
min_logins_th = transfers__rules_thresholds['min_account_activity_3m_th'].threshold;

percentage_login_time_of_day = Maths.safeDivision(event.count_logins_time_of_day_3m,
event.count_logins_3m);
```

Condition

```
percentage_login_time_of_day <= max_logins_time_of_day_th
and event.count_logins_3m >= min_logins_th
and event.count_sender_receiver_6m == 0
```

Explanation

[Transfers-FR15] - High Risk - Transaction is outside of usual time of day for this account and is to a new receiver (last 6 months)

Mark As

The outcome of this rule is: **high**

Transfers-FR18

Comment

Monthly limit on different senders per email address

Actions

- [Transfers-FR18]

Variables

```
min_distinct_sender_per_email_1m_th
=transfers__rules_thresholds['min_distinct_sender_per_email_1m_th'].threshold;
```

Condition

```
event.distinct_sender_per_email_1m >= min_distinct_sender_per_email_1m_th
```

Explanation

[Transfers-FR18] - High Risk - Count of distinct senders ({event.distinct_sender_per_email_1m}) for this email address ({event.sender_email}) in the last month is above {localvar.min_distinct_sender_per_email_1m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR19

Comment

Monthly limit on different senders per phone number

Actions

- [Transfers-FR19]

Variables

```
min_distinct_sender_per_phone_1m_th=transfers__rules_thresholds['min_distinct_sender_per_phone_1m_th'].threshold;
```

Condition

```
event.distinct_sender_per_phone_1m>= min_distinct_sender_per_phone_1m_th
```

Explanation

[Transfers-FR19] - High Risk - Count of distinct senders ({event.distinct_sender_per_phone_1m}) for this phone number({event.sender_phone}) in the last month is above {localvar.min_distinct_sender_per_phone_1m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR22

Comment

Multiple failed login attempts followed by transaction to a new added payee

Actions

- [Transfers-FR22]

Variables

```
failed_attempts_th = transfers__rules_thresholds["fr22_failed_attempts_th"].threshold;  
new_payee_transfers_th = transfers__rules_thresholds["fr22_new_payee_transfers_th"].threshold;
```

Condition

```
event.count_failed_logins_24h >= failed_attempts_th
and
event.count_new_payee_transfers_24h >= new_payee_transfers_th
```

Explanation

[Transfers-FR22] - Medium Risk - Transfers to new payees ({event.count_new_payee_transfers_24h}) exceeded {localvar.new_payee_transfers_th} for an account with multiple failed login attempts ({event.count_failed_logins_24h} > {localvar.failed_attempts_th}).

Mark As

The outcome of this rule is: **medium**

Transfers-FR23

Comment

Credentials Update or Profile Update followed by transferring funds that exceeds a percentage threshold of the account balance

Actions

- [Transfers-FR23]

Variables

```
percentage_th = transfers__rules_thresholds["fr23_percentage_th"].threshold;
```

Condition

```
(event.last_digital_event_type_24h == "credentials_update"
or
event.last_digital_event_type_24h == "profile_update")
and
event.sum_amount_per_sender_24h >= percentage_th * event.sender_account_balance
```

Explanation

[Transfers-FR23] - Medium Risk - Sender tried to transfer at least {localvar.percentage_th} x their account balance after having updated their credentials or profile.

Mark As

The outcome of this rule is: **medium**

Transfers-FR24

Comment

Number of different Senders per Device ID in 90 minutes exceeds threshold

Actions

- [Transfers-FR24]

Variables

```
min_distinct_sender_per_device_id_90m_th=transfers__rules_thresholds['min_distinct_sender_per_device_id_90m_th'].threshold;
```

Condition

```
event.distinct_sender_per_device_id_ >= min_distinct_sender_per_device_id_90m_th
```

Explanation

[Transfers-FR24] - High Risk - Count of distinct senders ({event.distinct_sender_per_device_id_}) for this Device Id in last 90 minutes is above {localvar.min_distinct_sender_per_device_id_90m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR25

Comment

Number of different Device IDs per Sender in 90 minutes exceeds threshold

Actions

- [Transfers-FR25]

Variables

```
min_distinct_device_id_per_sender_90m_th=transfers__rules_thresholds['min_distinct_device_id_per_sender_90m_th'].threshold;
```

Condition

```
event.distinct_device_id_per_sender_>=min_distinct_device_id_per_sender_90m_th
```

Explanation

[Transfers-FR25] - High Risk - Count of distinct devices ({event.distinct_device_id_per_sender_}) per Sender in last 90m is above {localvar.min_distinct_device_id_per_sender_90m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR27

Comment

Daily number of failed transactions per sender exceeds threshold

Actions

- [Transfers-FR27]

Variables

```
min_count_failed_trx_per_sender_24h_th=transfers__rules_thresholds['min_count_failed_trx_per_sender_24h_th'].threshold;
```

Condition

```
event.count_failed_trx_sender_24h >= min_count_failed_trx_per_sender_24h_th
```

Explanation

[Transfers-FR27] - High Risk - Count of failed transactions from this Sender in the last 24hours
({event.count_failed_trx_sender_24h}) is above {localvar.min_count_failed_trx_per_sender_24h_th}.

Mark As

The outcome of this rule is: **high**

Transfers-FR28

Comment

Decline transaction if email domain is gmail and contains a "_" or a "+"

Actions

- [Transfers-FR28]

Variables

N/A

Condition

```
StringUtils.substringAfterLast(event.sender_email ?? '', '@') ?? '' == 'gmail.com'  
and  
(StringUtils.contains(event.sender_email ?? '', '+') or  
StringUtils.contains(event.sender_email ?? '', '_'))
```

Explanation

[Transfers-FR28] - High Risk - Email domain is gmail and contains invalid characters ("_" or "+")

Mark As

The outcome of this rule is: **high**

Transfers-FR30

Comment

Distance between current and previous ip location in the last hour exceeds 1000 km (1 hour)

Actions

- [Transfers-FR30]

Variables

```
min_sender_distance_km_1h_th =  
transfers__rules_thresholds["min_sender_distance_km_1h_th"].threshold;  
  
dist =  
Location.distFromInKm(event.device_latitude_double,event.device_longitude_double,event.sender_prev_device_lat_1h,event.sender_prev_device_long_1h) ?? 0.0;
```

Condition

```
(event.sender_prev_device_lat_1h ?? 0.0 != 0.0) and (event.sender_prev_device_long_1h ?? 0.0 != 0.0)
and
dist >= min_sender_distance_km_1h_th
```

Explanation

[Transfers-FR30] - High Risk - Distance between previous and current IP location in the last hour is higher than {localvar.min_sender_distance_km_1h_th} km

Mark As

The outcome of this rule is: **high**

Transfers-FR32

Comment

Transaction from a device_id with login activity linked to multiple accounts, in the last 3 months

Actions

- [Transfers-FR32]

Variables

```
count_th = transfers__rules_thresholds['distinct_account_per_device_3m_th'].threshold;
```

Condition

```
event.distinct_account_login_per_dev >= count_th
```

Explanation

[Transfers-FR32] - Medium Risk - The number of logins with distinct account id per device ({event.device_id}) in the last 3m ({event.distinct_account_login_per_dev}) is above {localvar.count_th}

Mark As

The outcome of this rule is: **medium**

Transfers-FR33

Comment

Funds sent exceed daily average amount after adding a number of new payees that exceeds a certain threshold

Actions

- [Transfers-FR33]

Variables

```
ratio_th = transfers__rules_thresholds["fr33_ratio_th"].threshold;
count_th = transfers__rules_thresholds["fr33_count_th"].threshold;

ratio = Maths.safeDivision((event.sender_transaction_amount ?? 0) * 1.0,
event.avg_daily_amount_per_sender_1w);
```

```
formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
```

Condition

```
event.new_payees_24h > count_th
and
ratio >= ratio_th
```

Explanation

[Transfers-FR33] - High Risk - Transaction amount ({localvar.formatted_amount}) exceeds ({localvar.ratio_th} x) the sender's daily average after adding at least {localvar.count_th} new payees in the last 24 hours.

Mark As

The outcome of this rule is: **high**

Transfers-FR34

Comment

Number of inbound transactions in the last 24 hours exceed a certain threshold for an account with many new beneficiaries in the last 24 hours when using a VPN or Tor connection

Actions

- [Transfers-FR34]

Variables

```
sender_count_th = transfers__rules_thresholds["fr34_sender_count_th"].threshold;
transaction_count_th = transfers__rules_thresholds["fr34_transaction_count_th"].threshold;

is_suspicious_network =
  Revelock.existsAlertOfType(event.revelock_alerts , 'Tor connection') ?? false
  or Revelock.existsAlertOfType(event.revelock_alerts , 'VPN connection') ?? false;
```

Condition

```
event.count_inbound_24h >= transaction_count_th
and
event.distinct_receiver_per_sender_2 >= sender_count_th
and
is_suspicious_network
```

Explanation

[Transfers-FR34] - High Risk - Inbound transactions in the last 24 hours ({event.count_inbound_24h}) exceed threshold ({localvar.transaction_count_th}) for an account with {event.distinct_receiver_per_sender_2} new beneficiaries in the last 24 hours.

Mark As

The outcome of this rule is: **high**

Transfers-FR35

Comment

Ratio of funds received to funds sent is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- [Transfers-FR35]

Variables

```
ratio_th = transfers__rules_thresholds["fr35_ratio_th"].threshold;
total_th = transfers__rules_thresholds["fr35_total_th"].threshold;

ratio = Maths.safeDivision(event.sum_amount_per_receiver_7d, event.sum_amount_per_sender_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
event.sum_amount_per_receiver_7d >= total_th
and
event.sum_amount_per_sender_7d >= total_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

[Transfers-FR35] - High Risk - Ratio of funds received to funds sent falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

The outcome of this rule is: **high**

Transfers-FR36

Comment

Ratio of funds sent to funds received is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- [Transfers-FR36]

Variables

```
ratio_th = transfers__rules_thresholds["fr36_ratio_th"].threshold;
total_th = transfers__rules_thresholds["fr36_total_th"].threshold;

ratio = Maths.safeDivision(event.sum_amount_per_sender_7d, event.sum_amount_per_receiver_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
event.sum_amount_per_receiver_7d >= total_th
and
event.sum_amount_per_sender_7d >= total_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

[Transfers-FR36] - High Risk - Ratio of funds sent to funds received falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

The outcome of this rule is: **high**

Transfers-FR37

Comment

Ratio of transactions sent to transactions received is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- [Transfers-FR37]

Variables

```
ratio_th = transfers__rules_thresholds["fr37_ratio_th"].threshold;
count_th = transfers__rules_thresholds["fr37_count_th"].threshold;

ratio = Maths.safeDivision(event.count_per_sender_7d, event.count_per_receiver_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
event.count_per_receiver_7d >= count_th
and
event.count_per_sender_7d >= count_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

[Transfers-FR37] - High Risk - Ratio of transactions sent to transactions received falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

The outcome of this rule is: **high**

Transfers-FR38

Comment

Ratio of transactions received to transactions sent is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- [Transfers-FR38]

Variables

```
ratio_th = transfers__rules_thresholds["fr38_ratio_th"].threshold;
count_th = transfers__rules_thresholds["fr38_count_th"].threshold;

ratio = Maths.safeDivision(event.count_per_receiver_7d, event.count_per_sender_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
event.count_per_receiver_7d >= count_th
and
event.count_per_sender_7d >= count_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

[Transfers-FR38] - High Risk - Ratio of transactions received to transactions sent falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

The outcome of this rule is: **high**